



Problem Set 3

Due: 7 June 2024, 3:30 p.m.

Problem 1. A particle moves along a circle with radius R , so that the tangential component of its acceleration is constant. At $t = 0$ the velocity of the particle was equal to zero. At t_0 find

- (a) the magnitude of the normal component a_n of the acceleration,
- (b) the magnitude of the acceleration vector $\mathbf{a}$ as well as the angle the vector $\mathbf{a}$ forms with the position vector $\mathbf{r}$,

as functions of time.

(2 + 2 points)

Problem 2. A small metal ball is suspended on a string attached to the roof of a car. Find the angle that the string forms with the vertical direction, if the car:

- (a) moves with constant speed along a straight line,
- (b) moves with constant acceleration along a straight line,
- (c) slides without friction down a plane inclined at an angle α to the horizontal.

Solve the problem in an inertial frame of reference (clearly indicate the frame of reference you are solving the problem in). Sketch relevant free body diagrams.

(1 + 2 + 2 points)

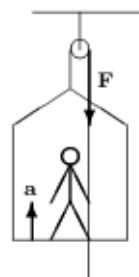
Problem 3. A particle with mass $m = 2$ kg is acted upon by force $\mathbf{F} = (4 \sin 2t, 6t - 12, -6e^{-3t})$ N (the numbers here are assumed to have correct units). Assuming initial conditions $\mathbf{r}(0) = (5, 2, -3)$ m and $\mathbf{v}(0) = (2, 0, 1)$ m/s find the velocity and position of the particle at any instant of time t .

(3/2 + 3/2 points)

Problem 4. A student of weight 320 N stands on a wooden bar of weight 160 N (see the figure), and pulls the rope downwards with force 250 N. Find

- (a) the acceleration of the student as he moves upwards,
- (b) the force he exerts on the bar.

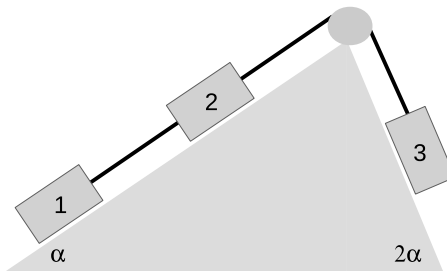
(3 + 1 points)



Problem 5. Three blocks with masses m_1 , m_2 , and m_3 are connected by massless strings and placed on planes inclined at the angles α and 2α , as shown in the figure below. The pulley is frictionless, and the coefficients of kinetic friction between blocks 1 and 2 and the surface are equal to μ_1 and μ_2 , respectively. There is no friction between block 3 and the incline.

- (a) Assuming that the system moves so that block 3 slides downwards accelerating, find the acceleration of the blocks and the tensions in all strings.
- (b) What condition (relating the masses and the coefficients of friction) needs to be satisfied, if the blocks are to move as assumed in part (a)?

(5 + 1 points)



Problem 6. A uniform rope with mass m and length l is placed on a horizontal table. One end of the rope is attached to a block with mass M resting on the same table. The other end of the rope (which remains always on the table) is pulled with a horizontal force of magnitude F , and the system moves with acceleration. The coefficient of kinetic friction between the block and the surface is μ_k , and there is no friction between the rope and the surface. Find the tension in the rope as a function of the distance from the block.

(5 points)

